CURRICULUM VITAE

TAPOSH KUMER SARKER, PH.D.

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EDUCATION

Doctor of Philosophy, Computer Science — **August 2026**
University of Texas at El Paso — *Texas, USA*

Master of Engineering, Computer Science and Technology — **July 2014**
University of Electronic Science and Technology of China — *Sichuan, China*

Bachelor of Science, Computer Science and Engineering — **June 2007**
Khulna University — *Khulna, Bangladesh*

RESEARCH INTERESTS

Cyber-Physical Systems (CPS), Edge AI, Resource-Constrained ML, Time-Sensitive Networking (TSN).

PH.D. DISSERTATION

Title: *Integrated Framework for TSN-Enabled OT Networks and Scalable Edge Computing to Enable Real-Time Feedback Loop.*
Supervisor: Dr. Deepak K. Tosh.
Committee: Dr. Shirley V. Moore, Dr. Jaime Acosta, Dr. Joshua Suetterlein (PNNL).
Summary: Proposed the Edge-Augmented Real-Time Industrial Control System (EA-RICS) to seamlessly converge Operational Technology (OT) and Information Technology (IT) networks. On the IT side, the research optimizes deep neural networks (DNN) using hardware-aware neural architecture search (HW-NAS) and supernets to guarantee time-bound anomaly detection at the edge. On the OT side, it utilizes programmable data planes (P4), network calculus, and Time-Sensitive Networking (TSN) to ensure determinism. Finally, an eBPF-based low-jitter control loop is established to connect the IT and OT environments.

PUBLICATIONS

Peer-Reviewed & Accepted

- **Taposh Kumer Sarker**, Deepak K. Tosh, Shirley V. Moore and Joshua Suetterlein. "Improving Admissibility of Time-Sensitive Flows in Industrial Operational Technology Networks." *The 2026 IEEE International Conference on Smart Grid Communications (SmartGridComm).* IEEE, 2026. Accepted.
- **Taposh Kumer Sarker** and Deepak K. Tosh. "Traffic Shaping to Traffic Engineering in Time-Sensitive OT Network", in *2025 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Toronto, Canada,* pp. 1-6.
- **Taposh Kumer Sarker** and Deepak K. Tosh. "Low latency and resilient industrial network with next generation SDN and PLC virtualization", in *MILCOM 2024-2024 IEEE Military Communications Conference (MILCOM), Washington, DC, USA,* pp. 1040-1045.
- T. Hossain, R. S. Shange, R. O. Ankumah, S. Fall, W. H. McElhenney, J. E. Quansah, **T. K. Sarker**, and J. J. Aun. "Prokaryotic Community Composition and Diversity in Sediments from Choctafaula, Uphapee, and Cubahatchee Creeks in the Lower Tallapoosa River Basin, Macon County, Alabama: An Observational Study." *Air, Soil and Water Research*, vol. 17, 2024.

Under Review

- **Taposh Kumer Sarker**, Deepak K. Tosh, Shirley V. Moore and Joshua Suetterlein. "From Best-Effort to Statistical Determinism: Optimizing Edge Inference Latency for Industrial Control Loops using eBPF." *IEEE Consumer Communications & Networking Conference 2027.*

TEACHING EXPERIENCE

Graduate Teaching Assistant **August 2021 - May 2022 & August 2023 - May 2024**
University of Texas at El Paso *Texas, USA*

CS 3331 - Advanced Object-Oriented Programming *Fall 2023, Spring 2024*

- Evaluated class tests and exams for a large cohort of over 100+ students, and served as a substitute lecturer to conduct classes whenever the instructor was away.
- Co-designed and evaluated comprehensive semester-long Java projects structured to directly reflect core course concepts, including UML modeling, OOP principles, design patterns, unit testing, file I/O operations etc.
- Directly mentored assigned student teams (from a cohort of ∼30 groups) through the software development lifecycle, conducting weekly progress assessments and architectural code reviews.

CS 1320 - Computer Programming for Scientists and Engineers *Fall 2021, Spring 2022*

- Supported a cohort of approximately 40 students by evaluating exams and providing real-time, hands-on assistance during in-class programming exercises.

RESEARCH EXPERIENCE

Research Collaborator (UTEP-PNNL Joint Research) **August 2025 - August 2026**
High-Performance Computing (HPC) Group, Pacific Northwest National Laboratory (PNNL)
Collaborated to resolve performance bottlenecks in coupled simulations of physics engines and AI surrogate models. In tandem with these workflows, developed a Hardware-Aware Neural Architecture Search (HW-NAS) framework utilizing Once-For-All (OFA) supernets and greedy coreset subsampling for edge anomaly detection using deep neural networks. This joint optimization approach successfully bounded worst-case inference execution latency, while maintaining acceptable accuracy for structural defects and fine-grained micro-texture anomalies in resource-constrained edge environments.

Graduate Research Assistant **August 2024 - August 2026**
TRUstworthy CYBER Systems (TRUCYBER) Laboratory, UTEP *El Paso, TX*
Conducted research on programmable data planes, network resilience, and Time-Sensitive Networking (TSN) with a focus on Operational Technology (OT) networks. Engineered P4 programmable data planes that autonomously detect link congestion and formulated mixed-criticality traffic engineering methodologies utilizing TSN shapers. Furthermore, developed a deadline-aware routing algorithm and tighter flow-admission mechanism utilizing network calculus and the Pay-Burst-Only-Once (PBOO) principle.

Graduate Research Assistant **June 2022 - May 2023**
The DeBlasio Lab, UTEP *El Paso, TX*
Investigated machine learning methodologies for the positive identification of Resident Space Objects (RSOs) utilizing distance-invariant hyper- and multi-spectral spectroscopy. Engineered custom Convolutional Neural Networks (CNNs), specifically adapting ResNet architectures with specialized kernels and strides to process raw Focal Plane Array (FPA) observations. Furthermore, investigated the application of Variational Autoencoders (VAEs) to leverage semi-supervised representation learning.

MENTORING EXPERIENCE

Academic and Research Mentor **2021 – 2026**
Department of Computer Science & Affiliated Labs, UTEP

- Directed 5 undergraduate student teams per semester in CS 3331 (Advanced OOP), guiding them through the end-to-end development and execution of semester-long software projects.
- In the TRUCYBER Lab, mentored undergraduate and master's students in configuring network emulation environments (Mininet, OMNeT++).
- In Dr. DeBlasio's lab, mentored an undergraduate collaborator working on the same research project, teaching him how to utilize High-Performance Computing (HPC) resources and train ML models within an HPC environment.

Technical Lead and Industry Mentor **2014 – 2021**
Professional Software Engineering Roles

- Mentored and onboarded numerous recent graduates and junior software engineers.
- Led code reviews and trained teams on Software Requirements Specification (SRS) requirements gathering and project management practices.

INDUSTRY EXPERIENCE

Solution Art Ltd. **Feb. 2015 – July 2021**
Senior Software Engineer, Tech Lead *Dhaka, Bangladesh*
Conducted requirements analysis, prepared Software Requirements Specifications (SRS), architected solutions (API, DB, Auth Schemes, Design Patterns), performed code reviews, and managed multiple projects:

- ERP software for the Bangladesh Coast Guard (Coast Guard Resource Planning and Management System - CGRPMS), a flagship project of the Bangladesh Govt. under the 'Digital Bangladesh' initiative. The tech stack included PHP, Laravel, ReactJS, MySQL, and Apache HTTP server. My role primarily involved requirements analysis, database design, and system integration.
- War room (OpsRoom) automation for the 'Operations Directorate' of the Bangladesh Navy. The tech stack primarily consisted of .NET (desktop GUI), Golang (Web API), and MySQL. I was involved in the design and implementation of the database and web API.
- ERP software for Chemist Laboratories Ltd (now Progoti ERP). The tech stack included .NET Core, ReactJS, and MySQL. My role involved requirements analysis, implementing OIDC-based authentication for multi-app solutions, and designing APIs for several modules.
- Lab automation and report generation, distribution, and verification system for the Bureau of Research, Testing & Consultation (BRTC). The internal part of the solution integrated all testing labs and streamlined report generation management, implemented using Golang and Java (JavaFX for GUI). The external part is a web application behind a public-facing Apache web server used by the public to search, view, and verify test reports (verify.ce.buet.ac.bd). The internal and external MySQL databases were deployed in a master/slave synchronization configuration, with the external slave being read-only.

Solution Art Ltd. **Jan. 2011 – Jul. 2012**
Software Engineer *Dhaka, Bangladesh*

- Implemented software features based on technical specifications for enterprise and government clients.
- Developed the the Secondary Education Sector Investment Program (SESIP) portal (sesip.gov.bd) for the Bangladesh Ministry of Education.
- Developed the 'Production' module for Mundipharma (Bangladesh) Pvt. Ltd.'s .NET ERP.

Sourcetop, Inc. **Jan. 2010 – Nov. 2010**
Drupal Developer *Dhaka, Bangladesh*

- Built web portals and websites utilizing Drupal CMS for US-based corporate clients as part of a remote team (Tech stack: Drupal, PHP, MySQL, JavaScript, jQuery, Apache).
- Served as the team's Git specialist (Git, Gitosis), managing repositories, resolving merge conflicts, and executing final code merging and testing.
- Developed web portals such as BCHD, Fora Financial, The Fiscal Times, Protocol Builder Pro, BHP Toolkit, and Launch Math.

Arena Mobile [Now Neomobile (DigitalGO Group)] **Aug. 2007 – Dec. 2009**
Python Developer *Dhaka, Bangladesh*

- Wrote and maintained Python scripts to analyze massive volumes of telecom data, aggregating metrics to determine user preferences and multimedia consumption patterns.
- Automated daily, weekly, and monthly data aggregation pipelines via cron jobs, generating Excel reports and automatically distributing them to management via email.
- Built customized multimedia web portals for telecom clients utilizing the Pylons web framework, jQuery, and MySQL.

FUNDING AND GRANTS

- **U.S. Department of Energy (DOE), Office of Science:** Research supported under Award #DE-SC0024352 for collaborative research with PNNL on AI and High-Performance Computing.
- **U.S. Department of Energy (DOE), Office of Fossil Energy (FE):** Research supported under Award #DE-FE0031744 for the development of resilient OT networks.
- **National Science Foundation (NSF):** Research supported under Award #2239609 for investigations into trustworthy cyber systems and programmable data planes.
- **United States Space Force (USSF):** Research supported via the University Consortium for Research Opportunity (UCRO) for applying ML to identify Unresolved Resident Space Objects (URSOs).

PRESENTATIONS

- **"Traffic Shaping to Traffic Engineering in Time-Sensitive OT Network"**
  2025 IEEE International Conference on Smart Grid Communications (SmartGridComm) Toronto, Canada | 2025
- **"Low latency and resilient industrial network with next generation SDN and PLC virtualization"**
  IEEE Military Communications Conference (MILCOM 2024) Washington, DC, USA | 2024
- **"Innovative Analysis of Spectro-Temporal Signatures using Machine Learning for Ground-Based Remote Sensing of Unresolved Resident Space Objects"**
  4th Annual VOLTRON Meeting on Space-Domain Awareness, UTEP El Paso, TX | Feb. 2023

HONORS, AWARDS & ACADEMIC SERVICE

Peer Reviewer, IEEE Consumer Communications & Networking Conference 2026 January 2026
Anita Mochen Loya Fellowship, University of Texas at El Paso August 2021 - May 2022
Chinese Govt. Scholarship, University of Electronic Science and Technology of China August 2012 - July 2014
Volunteer, Bangladesh Open Source Network (BdOSN) 2008 - 2012